

IKHWAN MAULANA

AI Engineer | Machine Learning | Computer Vision | Python

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in machine learning, computer vision, deep learning, and AI-powered application development. Experienced in building end-to-end AI solutions using Python, PyTorch, TensorFlow/Keras, YOLO, OpenCV, Streamlit, and scikit-learn, spanning data preprocessing and model training to evaluation and production deployment.

Possesses a strong passion for engineering practical AI systems that solve real-world problems, with domain expertise in object detection, image classification, recommendation systems, regression modeling, and intelligent software integration. Accustomed to managing large-scale datasets, conducting empirical model architecture benchmarking, evaluating precision metrics, and integrating trained weights into production-ready software.

TECHNICAL SKILLS

- **Programming Languages:** Python, Golang, PHP, JavaScript (ES6+), SQL, Java, HTML5, CSS3
- **AI, Machine Learning & Deep Learning:** Supervised Learning, Deep Neural Networks (CNNs), Model Training & Fine-Tuning, Hyperparameter Optimization, Transfer Learning, Model Evaluation Metrics (mAP50, mAP50-95, Precision, Recall, F1-Score)
- **Computer Vision & Image Processing:** Object Detection (YOLOv8, YOLOv11s), Image Classification, Image Segmentation, OpenCV, Color Histogram & K-Means Clustering, Laplacian Texture Variance, Dataset Annotation & Augmentation (Roboflow)
- **Recommender Systems & NLP:** Content-Based Filtering, TF-IDF (Term Frequency-Inverse Document Frequency) Vectorization, Cosine Similarity Matrix, Text Preprocessing, Metadata Feature Extraction
- **AI Frameworks & Edge Deployment:** PyTorch, TensorFlow, Keras, Ultralytics YOLO, Scikit-Learn, TensorFlow Lite (TFLite Quantization & On-Device Mobile Inference), ONNX, Google Gemini API
- **Backend & Web Engineering:** PHP Native (MVC Architecture), Golang, RESTful API Development, Streamlit, MySQL (Relational Schema Design, Indexing, Query Optimization), AJAX / Fetch API, Leaflet JS Geolocation
- **Mobile & Frontend Development:** Flutter (Dart • Android On-Device AI Integration), Tailwind CSS, Bootstrap, Responsive UI/UX Design
- **Data Science & Developer Tools:** Pandas, NumPy, Matplotlib, Seaborn, Git, GitHub, Docker, Linux/Unix Shell, Postman, Jupyter Notebook, Google Colab

SELECTED AI & MACHINE LEARNING PROJECTS

Black Soldier Fly (BSF) Larva Growth Stage Detection & Classification System (Bachelor's Thesis)

Python | YOLOv11s | K-Means Clustering | Laplacian Variance | TensorFlow Lite | Flutter

- Developed an end-to-end computer vision system to detect and classify 4 distinct BSF larva growth stages (Young Larva, Mature Larva, Prepupa, Pupa) to provide automated harvest timing guidance, replacing subjective manual visual observation.
- Curated and annotated an extensive dataset of 21,974 images (28,388 annotations) across 4 growth classes, including custom preprocessing and data augmentation pipelines.
- Fine-tuned a YOLOv11s model for real-time object detection and classification; extracted dominant color features via K-Means Clustering and surface texture variations using Laplacian Variance.
- Formulated a rule-based inference pipeline to compute the developmental stage proportions from detection outputs, generating automated harvest schedule recommendations.
- Achieved average metrics of 0.9723 precision, 0.8315 recall, 0.8720 mAP50, and 0.6759 mAP50-95, with peak performance on the Prepupa stage (0.9950 mAP50).
- Converted the trained weights to TensorFlow Lite and deployed on-device within an Android mobile application built with Flutter for rapid, offline-capable field inference and visual analysis.

AI-Powered Movie Recommender System

Python | Streamlit | Content-Based Filtering | TF-IDF | Cosine Similarity

- Engineered an intelligent movie recommendation engine utilizing Content-Based Filtering over movie metadata (genres, keywords, overview).
- Implemented TF-IDF (Term Frequency-Inverse Document Frequency) feature vectorization and applied Cosine Similarity metrics to identify and rank similar films.
- Built an interactive Streamlit web application featuring real-time title search, genre and actor filtering, detailed movie profiles, dynamic poster fetching, and watchlist functionality.
- Integrated external movie metadata APIs and dynamic poster retrieval to deliver a seamless, cinematic user recommendation experience.

AI-Based Image Classification Application

Python | TensorFlow/Keras | Streamlit

- Developed an interactive image classification web application powered by pre-trained deep convolutional neural networks.
- Engineered automated image preprocessing and input dimension standardization prior to model inference.
- Integrated trained model pipelines into a Streamlit interface for instantaneous prediction, complete with file format validation and channel conversion.
- Designed a responsive, intuitive interface tailored for seamless accessibility by non-technical end-users.

AI-Powered Web Applications

PHP | MySQL | JavaScript | Tailwind CSS | Python | Gemini API

- Engineered web applications utilizing structured MVC architecture and normalized relational MySQL database schemas.
- Implemented secure authentication, CRUD data management, interactive admin dashboards, and asynchronous communication via Fetch API/AJAX.
- Integrated external Large Language Model capabilities via Google Gemini API into web application workflows.
- Designed mobile-responsive interfaces using Tailwind CSS prioritizing usability, accessibility, and clean maintainable code.

Malaka Jaya Digital MSME Community Web Portal

PHP Native | MySQL | JavaScript | Leaflet JS | PHPMailer | Gemini API

- Architected a community enterprise showcase and product catalog platform featuring a three-tier role system (Admin, Merchant, Resident), enabling direct customer ordering via WhatsApp.
- Developed public discovery pages including categorized MSME directories, product galleries, merchant location maps, community news publishing, and global keyword search.
- Implemented role-based access control with secure email OTP verification (PHPMailer) and password recovery flows.
- Constructed an Admin Dashboard for merchant onboarding verification, news management with Gemini AI writing assistance, user administration, and real-time email notifications.
- Built dedicated Merchant Dashboards displaying sales metrics and full product catalog CRUD with photo uploads.
- Integrated an interactive AI assistant chatbot (Google Gemini API) accessible across all portal pages for citizen inquiries.
- Designed a 10-table normalized relational database schema covering users, merchants, products, articles, and audit logs.

Student Academic Grading System — SMP Islam Al-Amin

PHP Native | MySQL (mysqli) | HTML/CSS | JavaScript

- Engineered a role-gated academic grading system (Admin, Teacher, Student) using PHP Native MVC and secure session authentication.
- Built student and teacher master data management modules for administrators with automated credential generation and search indexing.
- Developed teacher grading interfaces with dynamic weight calculations (Assignments 30%, Midterm 30%, Final 40%), real-time client-side grade recapitulation, upsert database logic, and 0–100 score validation.
- Created filterable grade summary tables, printable report cards, and student portals for personal academic record inspection.
- Developed administrative reporting modules with class-level filters, CSS print stylesheets (@media print), and TSV-based Excel export.
- Designed a relational schema linking users, students, teachers, and grades with automated pass/fail thresholding (≥ 70 = Pass).

SOFTWARE ENGINEERING EXPERIENCE & COMPETENCIES

- Engineered full-stack applications from relational database design and backend business logic to frontend implementation and production deployment.
- Experienced in diagnosing software runtime environments, resolving package dependency conflicts, resolving model compatibility issues, and configuring GPU/CUDA acceleration.
- Applied modular programming paradigms to cleanly decouple data ingestion, business logic, model inference, and presentation layers.
- Proficient in designing normalized MySQL database schemas, architecting RESTful APIs, executing Git/GitHub version control workflows, and deploying Python-based AI applications.

EDUCATION

Bachelor of Computer Science (S.Kom) in Informatics Engineering | Universitas Gunadarma — Bekasi, Indonesia

Faculty of Industrial Technology / Department of Informatics • NPM: 50422698 | GPA: 3.78 / 4.00 (Class of 2026)

Relevant Coursework: Artificial Intelligence, Machine Learning, Computer Vision, Software Engineering, Data Science, Database Systems

Computer and Network Engineering (TKJ) | SMK Islam Al-Amin

Graduated 2022 • Focus: Computer Networking, Hardware Maintenance, Server & Network Administration

CERTIFICATIONS & TECHNICAL TRAINING

- **Programmer** — National Professional Certification Board (BNSP) / LSP (2026)
- **Oracle for Intermediate** — Universitas Gunadarma (2025, 2-Month Intensive)
- **Golang for Intermediate** — Universitas Gunadarma (2025, 2-Month Intensive)
- **Programmer Course** — Universitas Gunadarma (2025, 1-Week Technical Bootcamp)
- **Wireless LAN Installation** — Universitas Gunadarma (2024)
- **Application Development Design** — Universitas Gunadarma (2024)

CONFERENCES & SEMINARS

- **Tech Talk: "Designing Future with Back End in Tech Industry"** — Google Developer Student Clubs (GDSC), Universitas Gunadarma (2024)
- **National Symposium: "Synergy of University and Industry Collaboration: Software and Hardware for Artificial Intelligence"** — Universitas Gunadarma (2025)

ORGANIZATIONAL EXPERIENCE

- **Head of Entrepreneurship & Skills Division** — Student Council (OSIS), SMK Islam Al-Amin (2021)
Led student entrepreneurship initiatives, organized hands-on technical workshops, and coordinated cross-functional student project teams.

ADDITIONAL INFORMATION

- **Languages:** Indonesian (Native), English (Intermediate / Professional Working Proficiency)
- **Interests:** Artificial Intelligence, Machine Learning, Computer Vision, Generative AI, Recommender Systems, Software Engineering, Cloud & Mobile Deployment